



Mechanical energy storage with focus on hydropower – status and future development



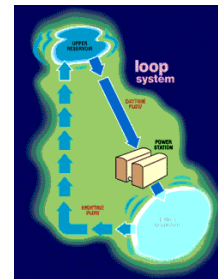
NATURHISTORISK MUSEUM
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***Atle Harby, SINTEF Energy Research and
Centre for environmental design of renewable energy - CEDREN***



Hydropower technology



Environmental impacts of hydropower



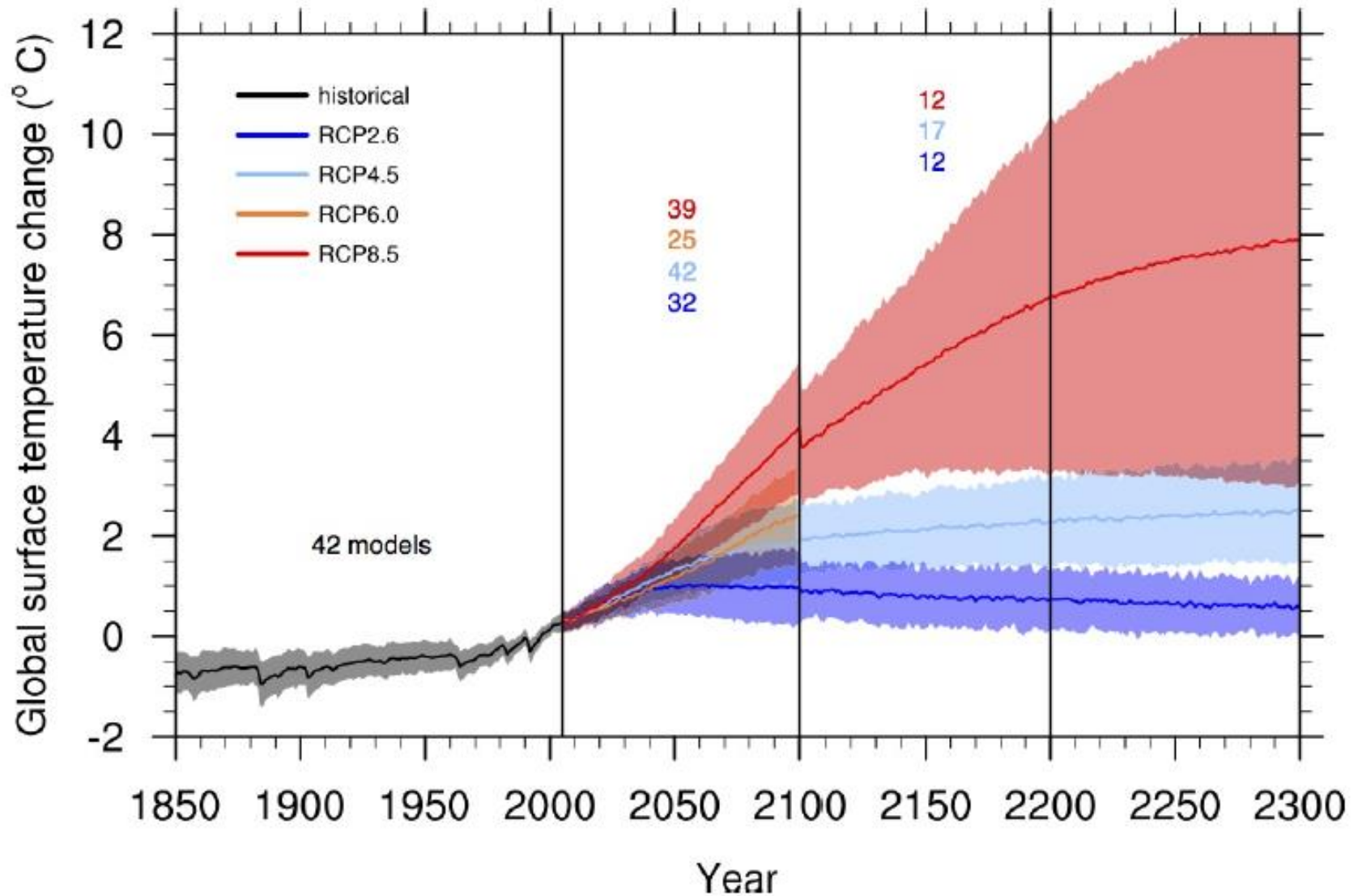
Environmental impacts of wind power and power transmission



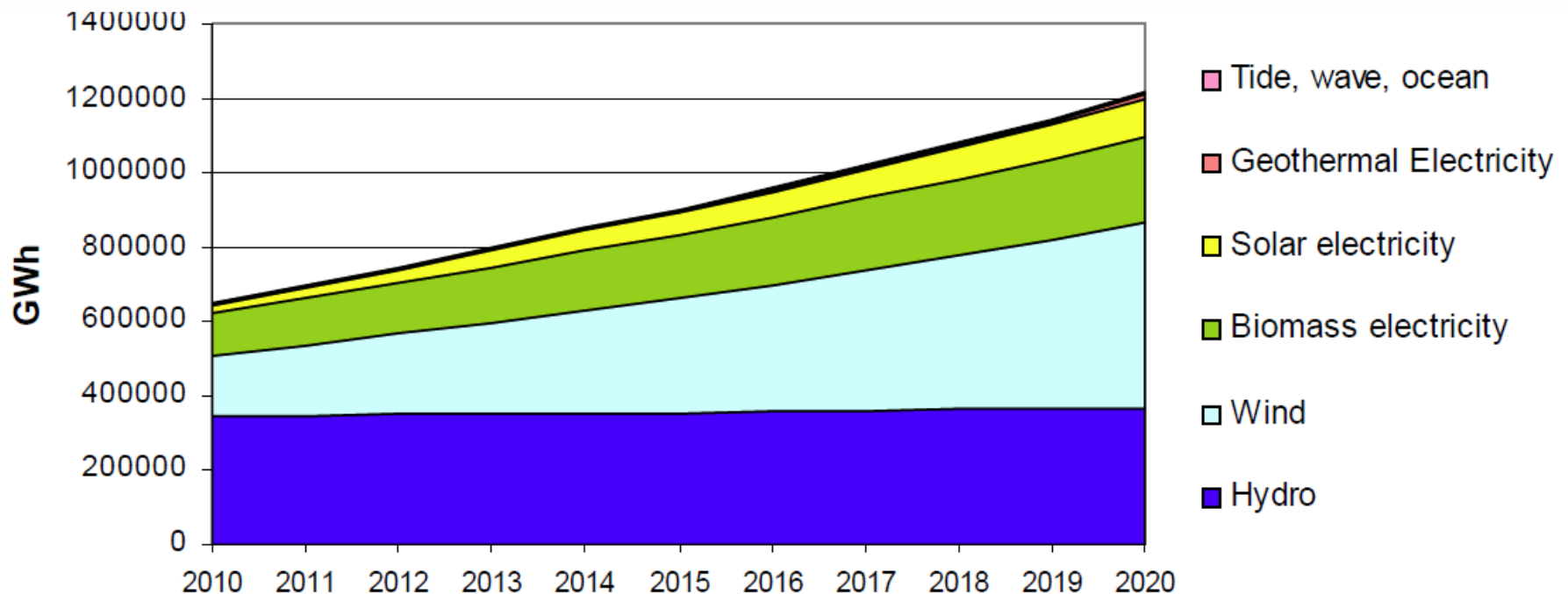
How to reconcile energy and environment policy?



The Background – Climate Change



Towards 2020 – Implementation of the RES-directive



RES generation from **632** TWh in 2010 to **1152** TWh in 2020

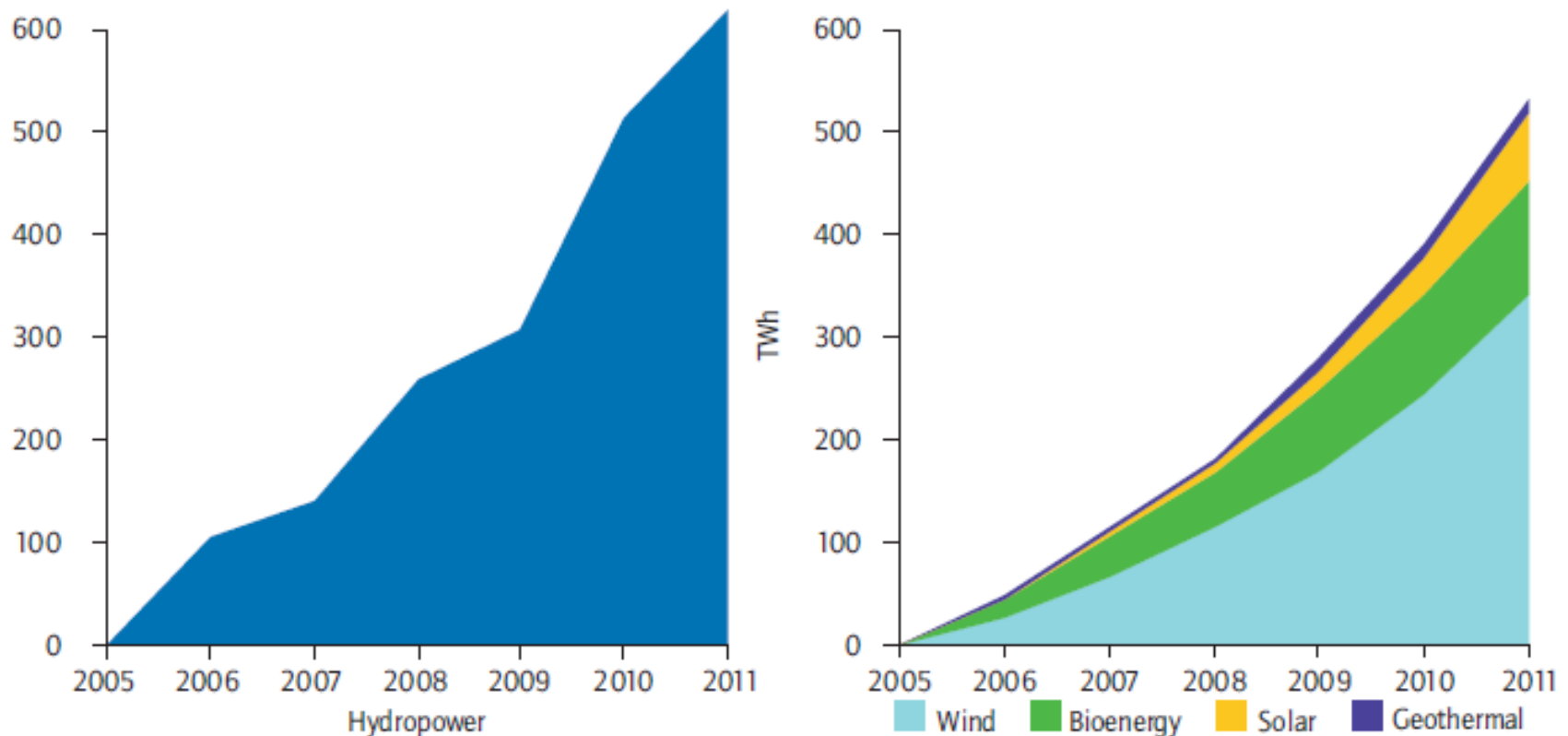
Largest increase in Wind - ca 120 GW and 305 TWh

Also rapid increase in Solar PV - ca 65 GW og 100 TWh

➔ Increase of **non-dispatchable** power generation (wind, solar PV)

A surprising fact found in IEA Roadmap cont.

Figure 3: Electricity generation from recent additions to hydropower (left) and other renewables (right)



Source: IEA, 2012b.

Energy scenarios

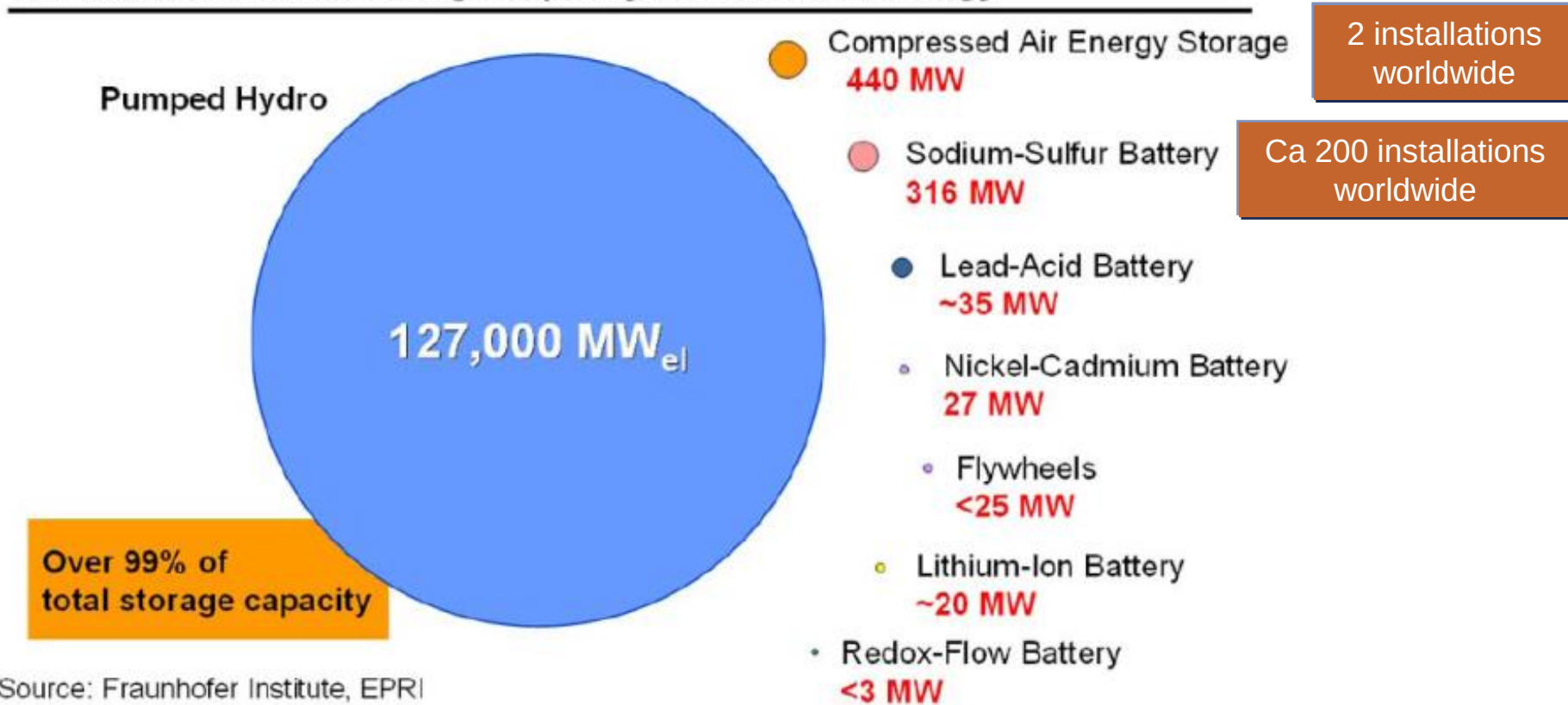


- ❑ Transmission and distribution infrastructure
- ❑ Energy storage technologies
- ❑ Demand side management
- ❑ Improved forecasting of resource availability

Maybe as much as 340 TWh of storage volume and 150 GW of balancing capacity needed in Europe by 2050

Installed Energy Storage capacity

Worldwide installed storage capacity for electrical energy



*Worldwide installed rated power of storage facilities for **electrical energy**.
Such power level can be sustained for up to several hours or shorter*

Mechanical storage

Hydro



- Operates typically on weeks to hours
- Many applications for both energy and storage
- World-wide potential

Compressed air



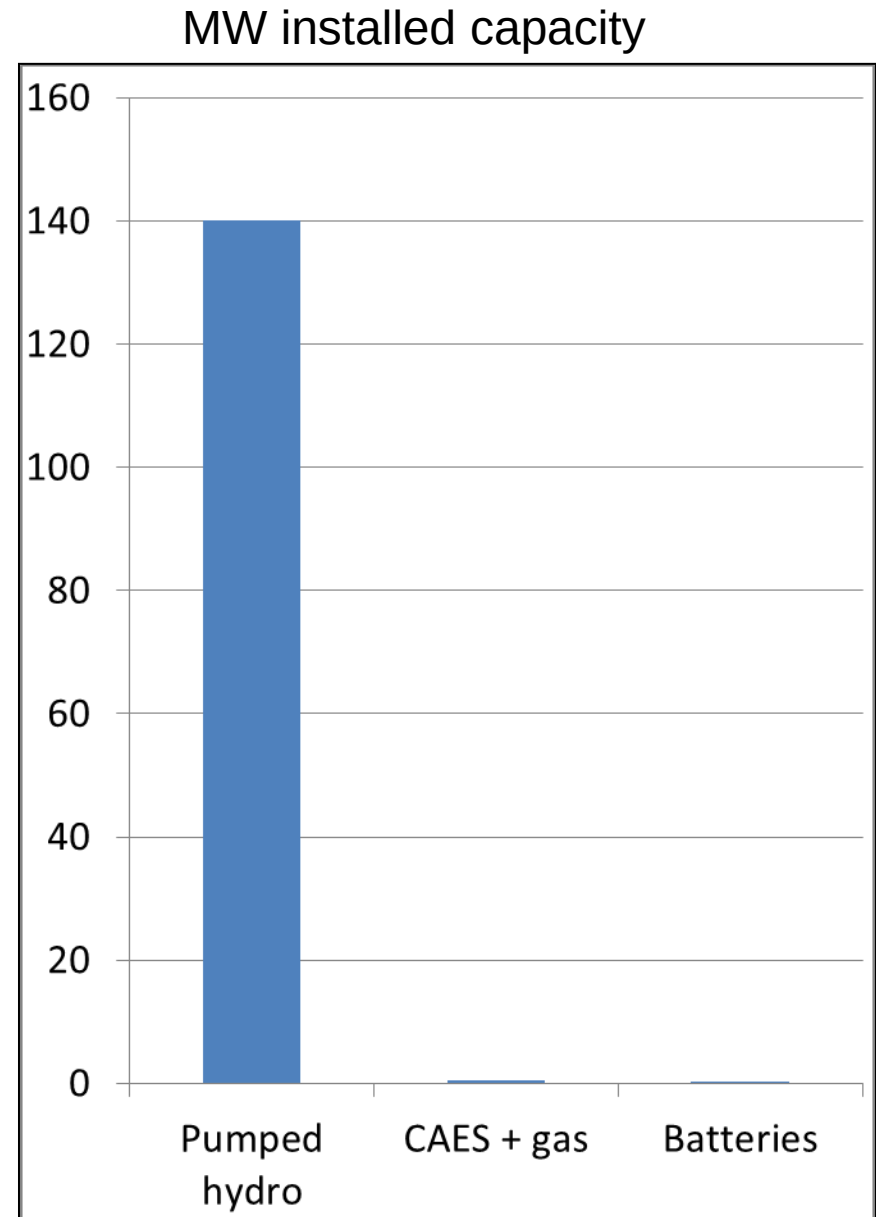
- Operates typically on hours
- Two commercial energy storage plants
- Need for more research

Flywheels



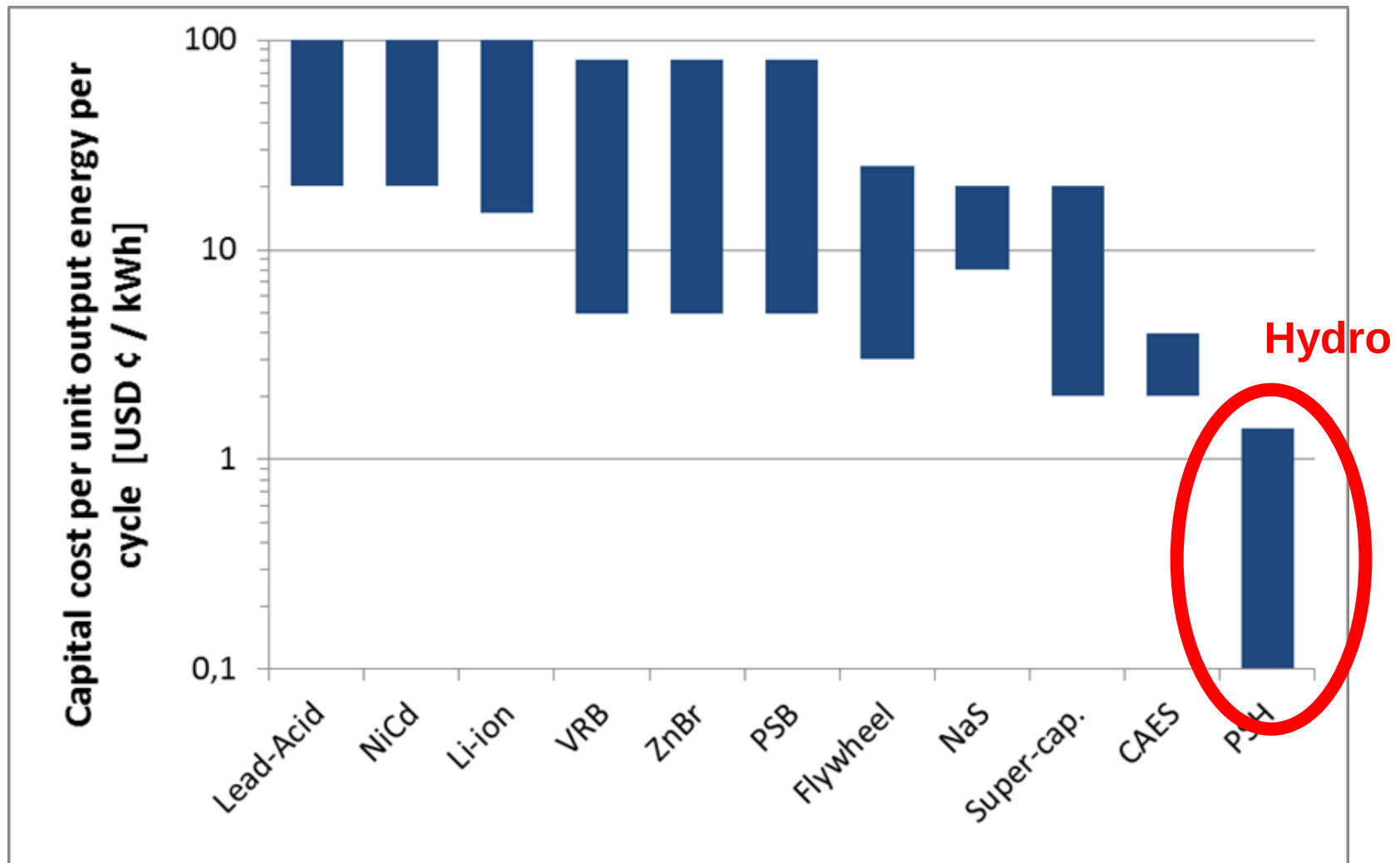
- Operates typically on seconds to minutes
- Used a lot in many other sectors
- Few large-scale energy storage applications

World electric energy storage installed capacity 2012



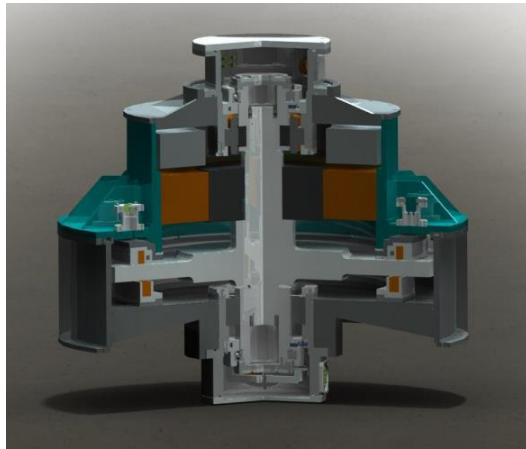
Capital costs per cycle:

Log scale

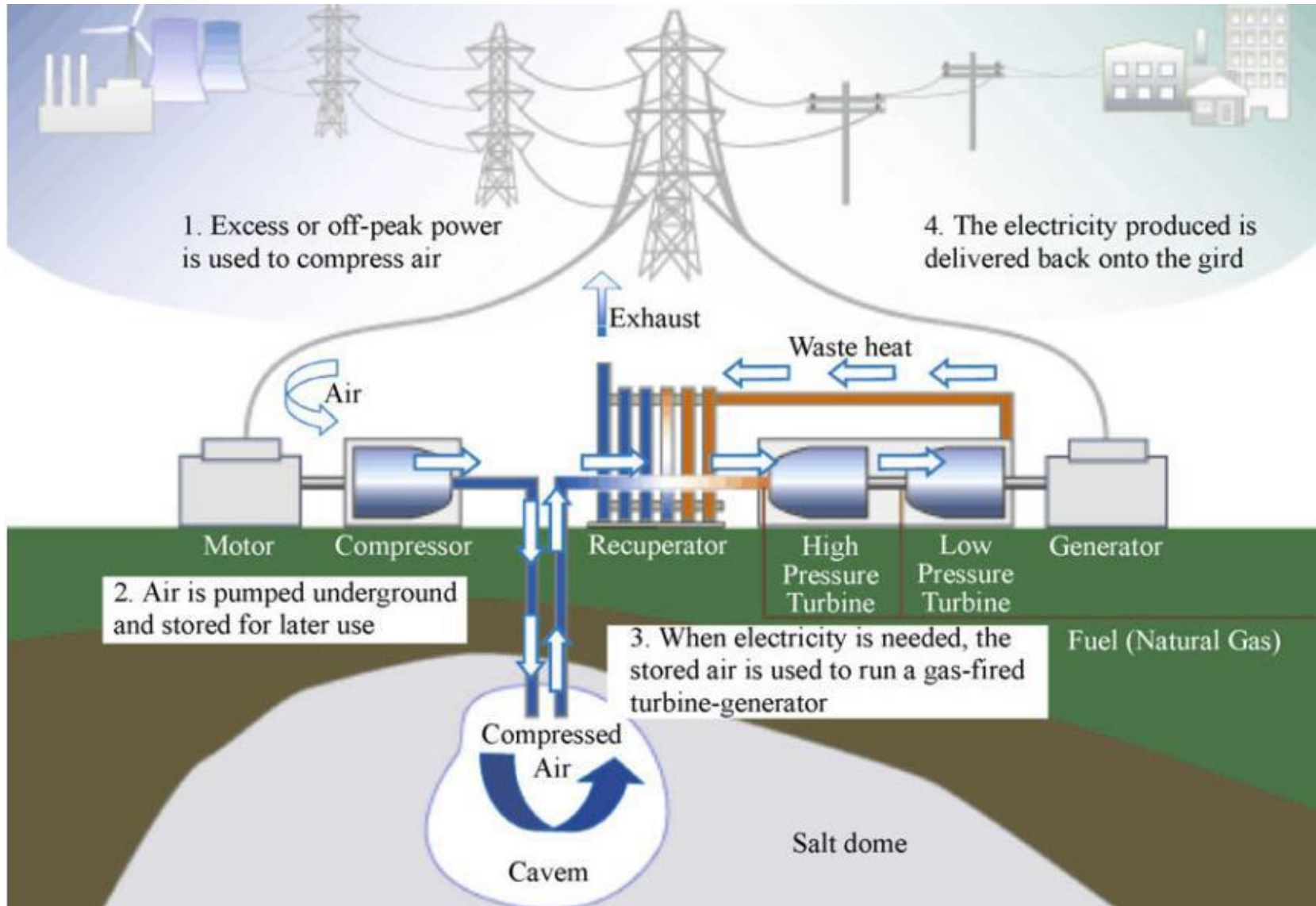


Flywheels – kinetic energy storage

- Rotating mass
- High power and energy density
- Very fast regulation
- A large number of life cycles
- Transportation
 - Electric and hybrid cars, buses
 - Light trains, trams, underground
 - Ferries
- Power system services
 - Grid stability
 - Frequency regulation
 - Voltage support
- Industry
 - Uninterrupted power supply
 - Cranes and elevators

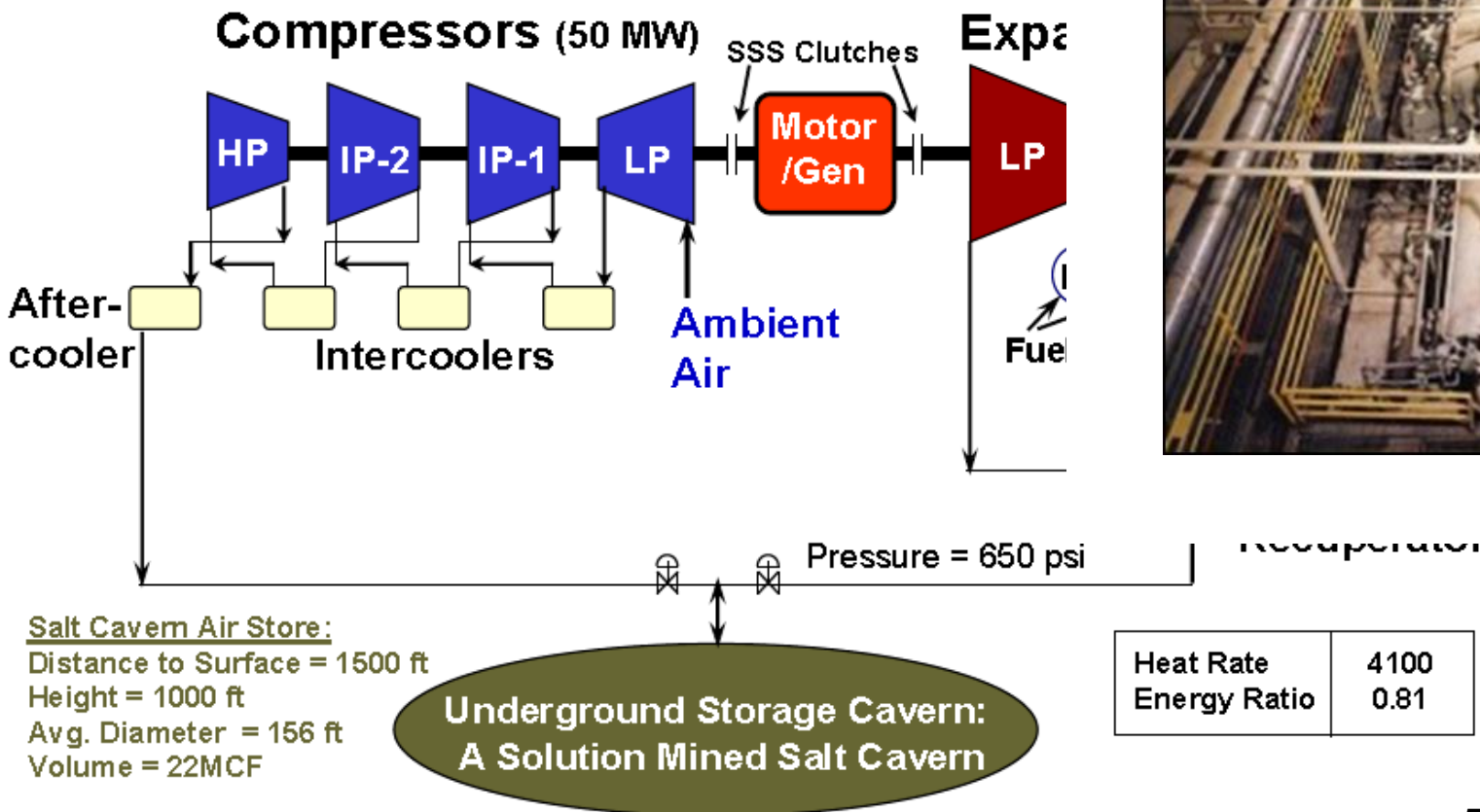


Compressed Air Energy Storage (CAES)



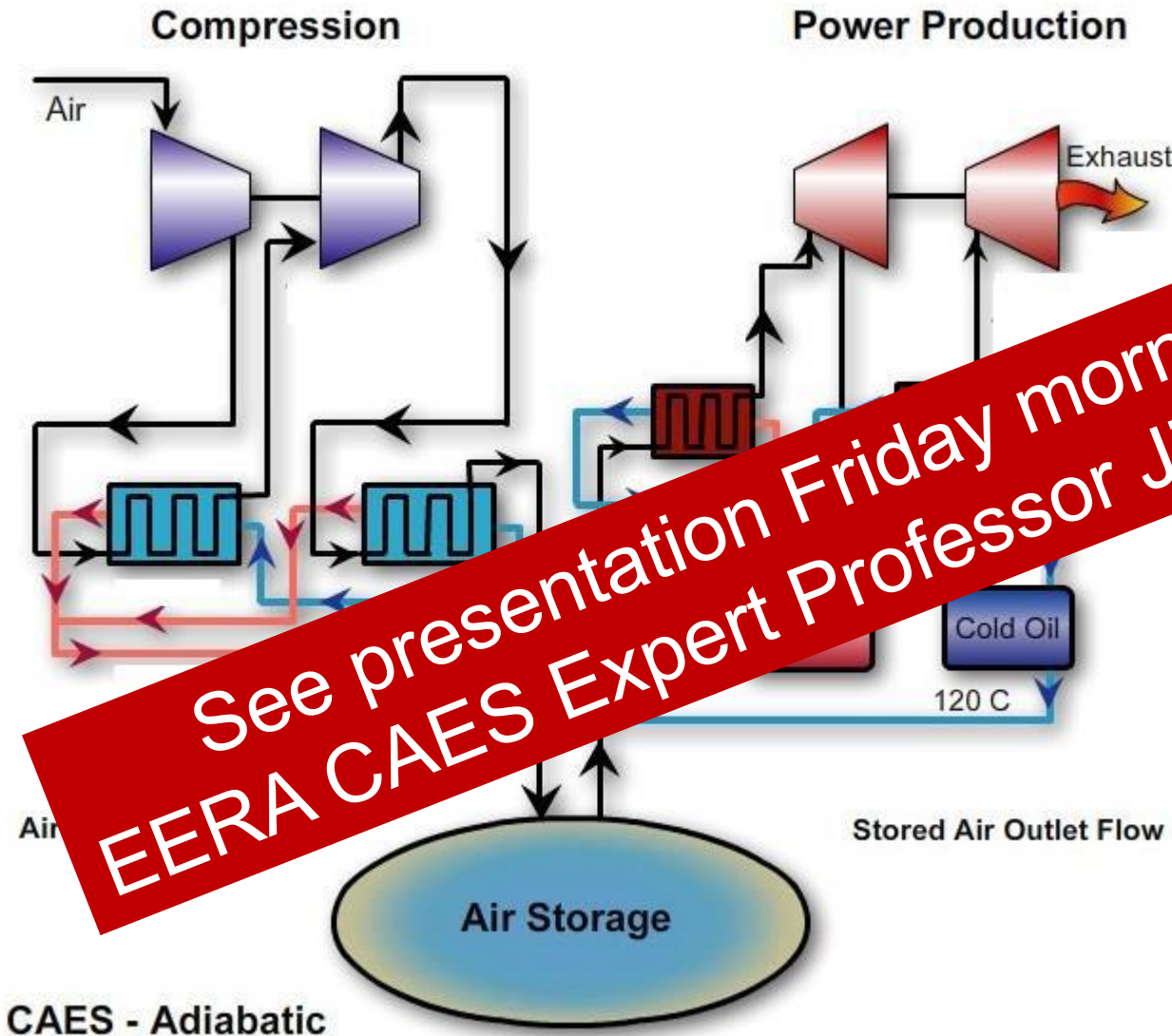
From Luo and Wang

McIntosh CAES plant (one of two commercial plants)



From Luo and Wang

Adiabatic CAES

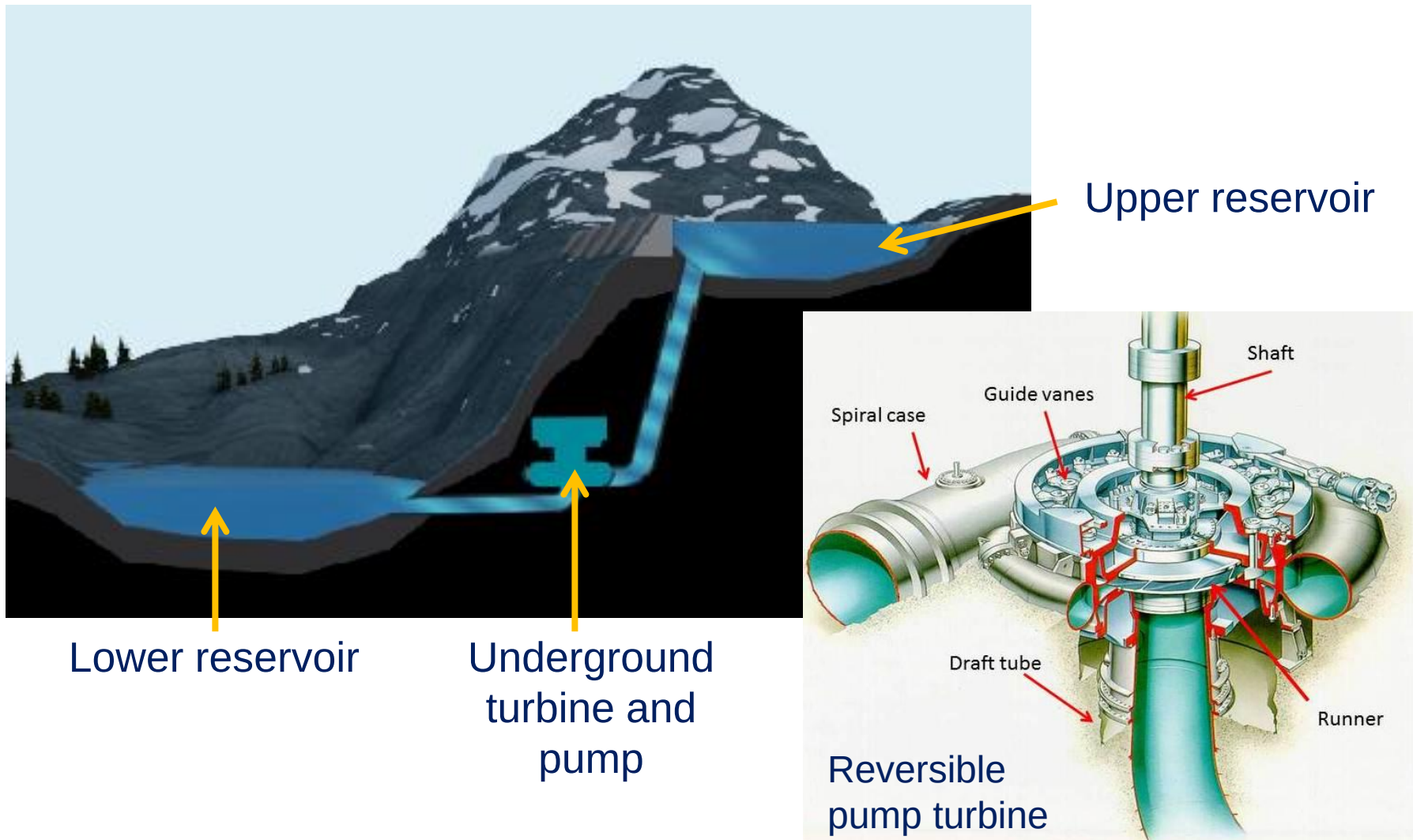


See presentation Friday morning by
EERA CAES Expert Professor Jihong Wang

Small scale
CAES:
Competing
with
batteries

From Luo and Wang

Pumped Storage Hydropower principle



Afourer, Marocco



Photo: Alstom

Nant de Drance, Switzerland

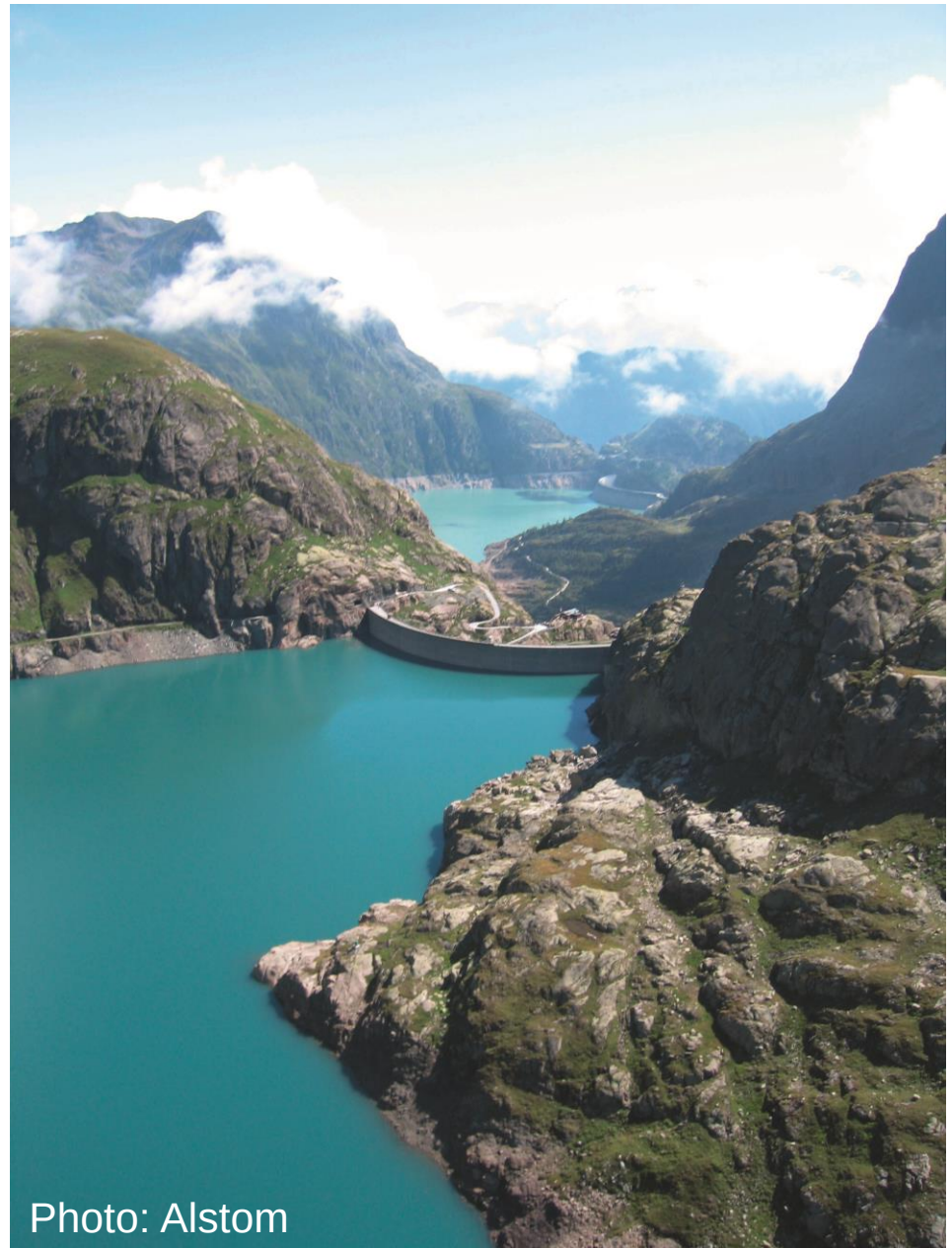


Photo: Alstom

Goldisthal, Germany



Limberg II, Austria



Grand Maison, France



1 800 MW turbine mode
1 275 MW in pump mode

Courtesy from Rioual, EdF



Vianden, Luxemburg

Upper reservoir

Lower reservoir:
The river

Photo: Alstom

Under Operation
Project stage

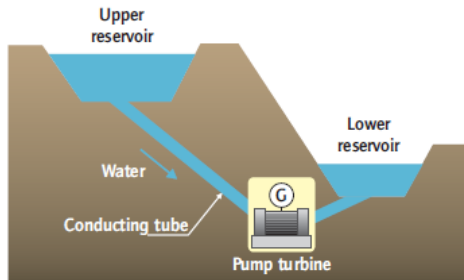
22 GW, 22 GW, 45 GW, 2 GW, 24 GW, 30 GW, 2 GW

Courtesy from Riquel, EdF

CEDREN Centre for Environmental Design of Renewable Energy  

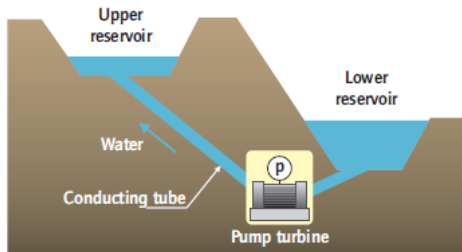
International PSP – Some facts

Existing capacity worldwide: 140 GW (2011)
42 GW added since 2005 (7 GW/year)
Projections 2050: 500-700 GW



Leading PSP regions

EU	45 GW
Japan	30 GW
China	24 GW
USA	20 GW

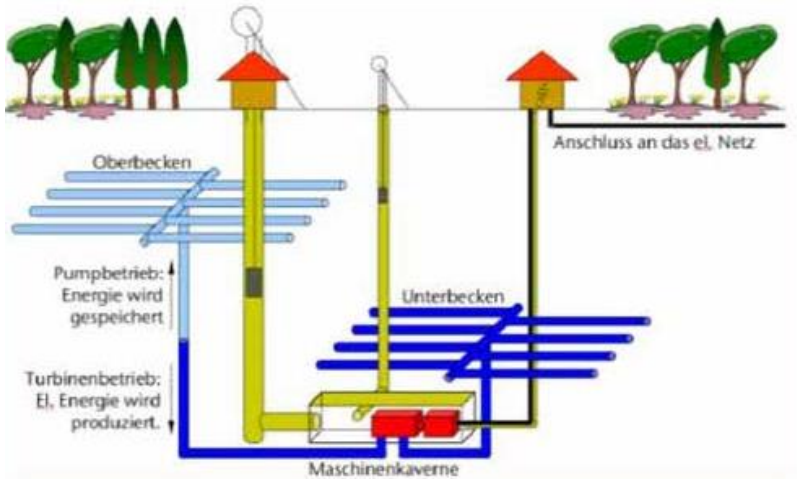


Storage:

Spain	17 PSP	1500 GWh
Switzerland	16 PSP	369 GWh
France	9 PSP	184 GWh
Austria	15 PSP	125 GWh

New technologies – pumped storage

- Underground PSP



- Sub-sea PSP
- Artificial island PSP
- Retrofitting reservoir hydro
- Variable speed reversible pump turbines

- Salt water PSP



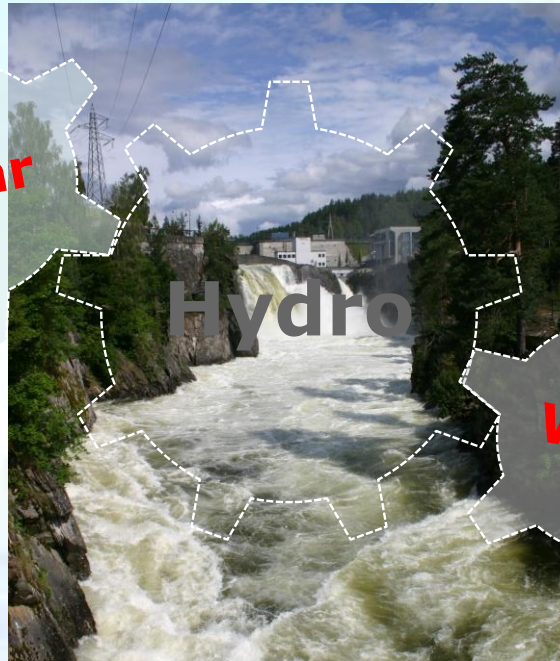
Okinawa PSP - Japan

- No access to lakes
- Scarce water resources
- Isolated grids (islands)
- Extra maintenance (salt)



Solar

Hydropower – supporting other renewables



Wind



CEDREN studies in 2011 and 2012: How can Norway contribute?



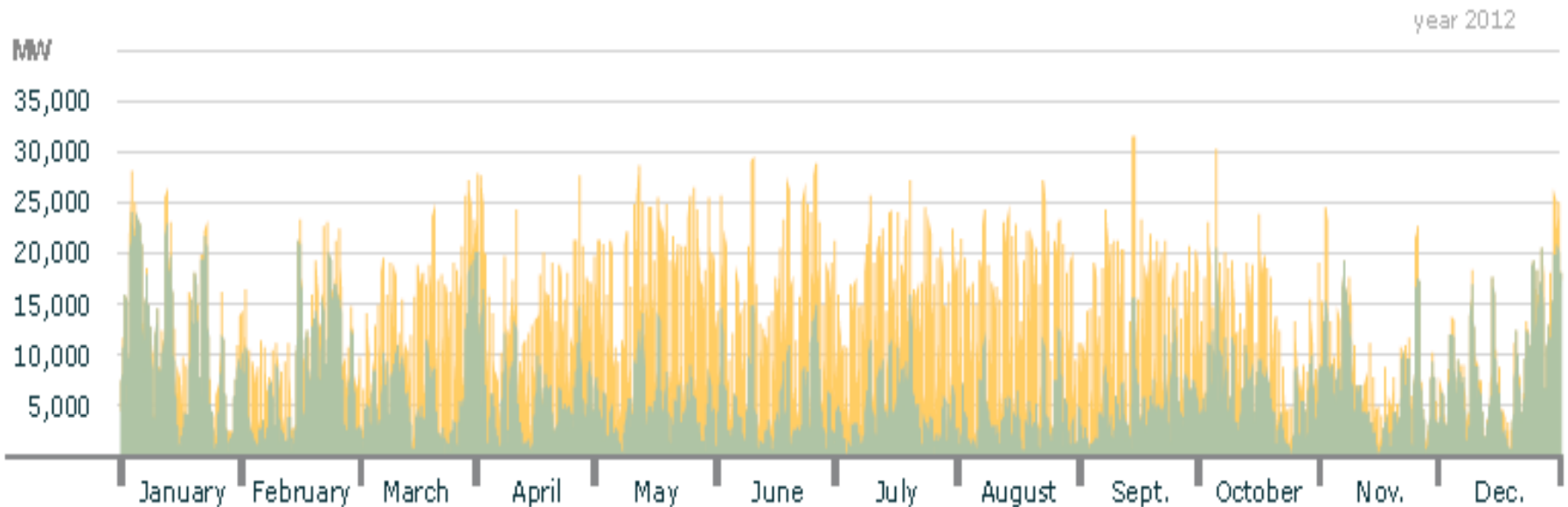
Main problem with Wind and Solar Power:

- Intermittency
- Highly variable output
- Low predictability
- Non-Dispatchable



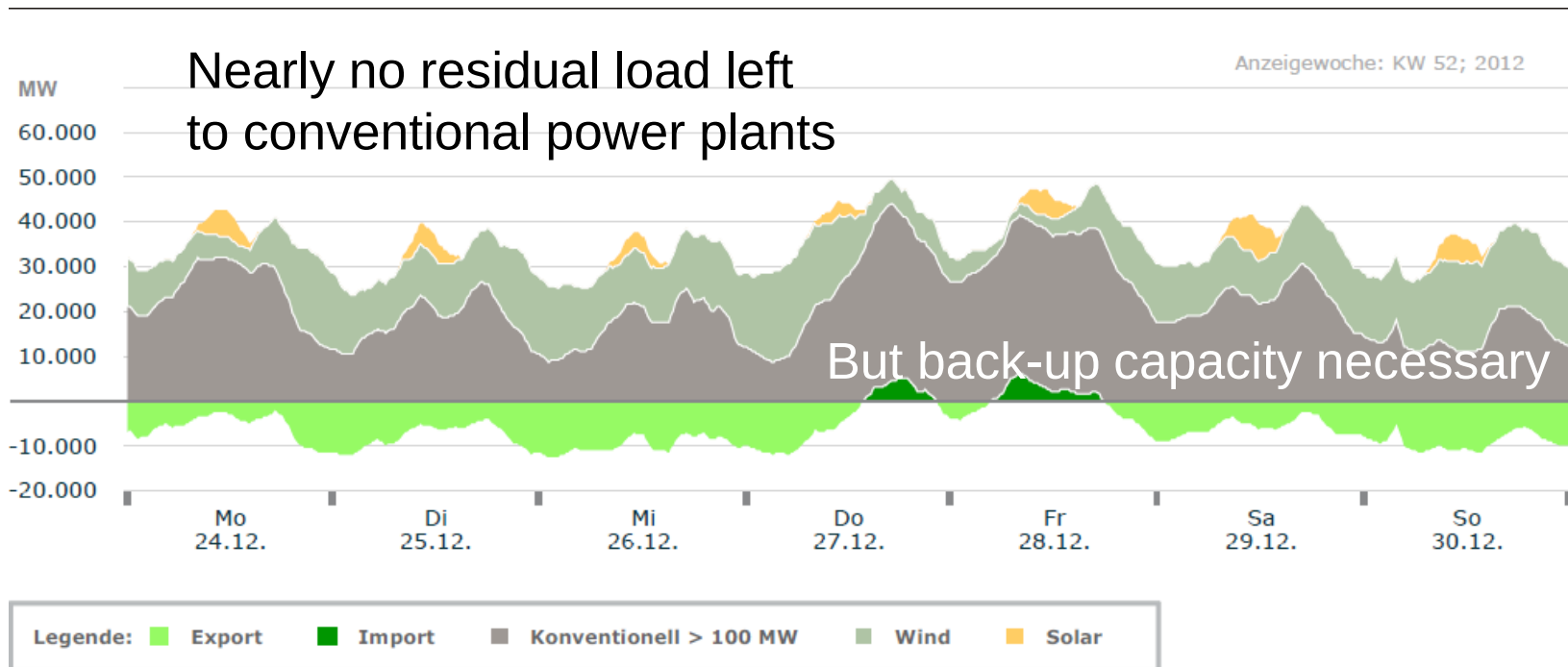
Actual production solar and wind

(Germany 2012)



RES challenges

High Wind power production

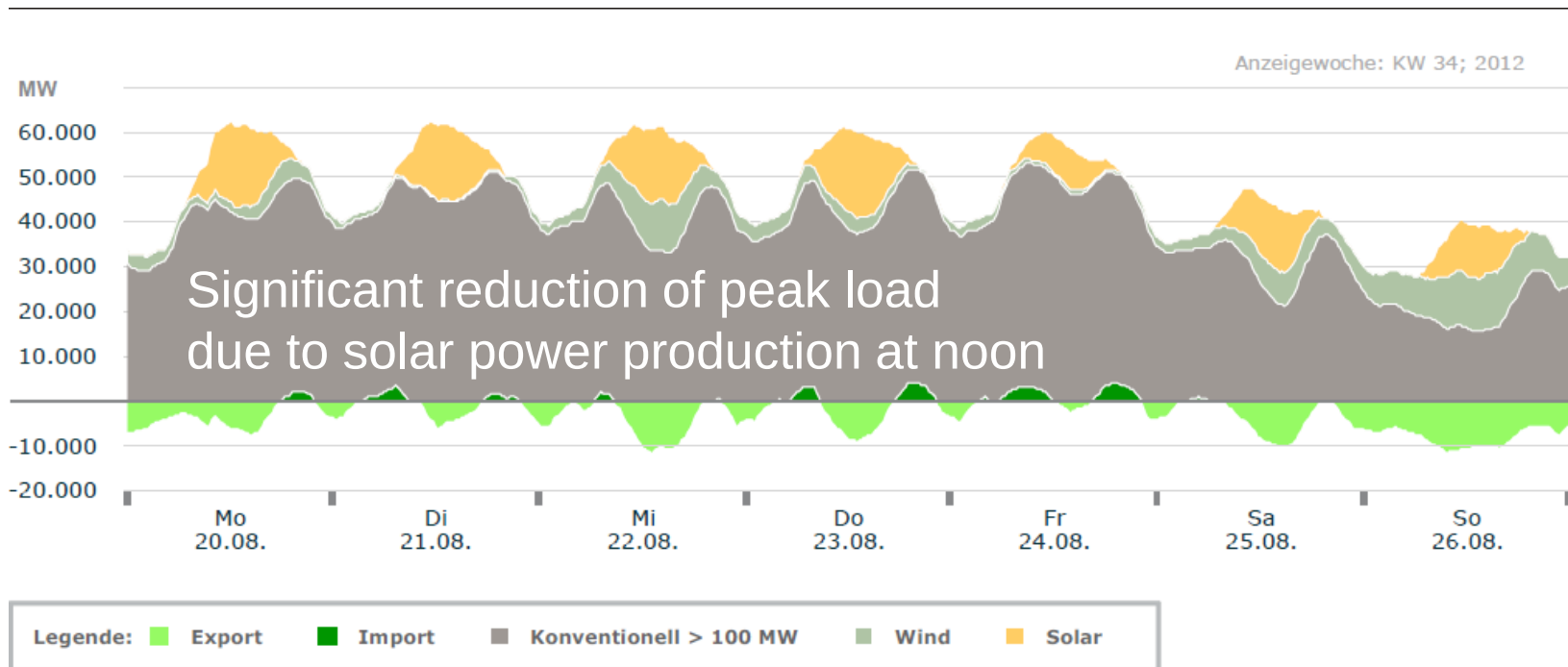


Source: Burger, B. "Electricity production from solar and wind energy in 2012",
Fraunhofer ISE, presentation, February, 2013

URL: <http://www.ise.fraunhofer.de/en/renewable-energy-data>

RES challenges

High Photovoltaic production

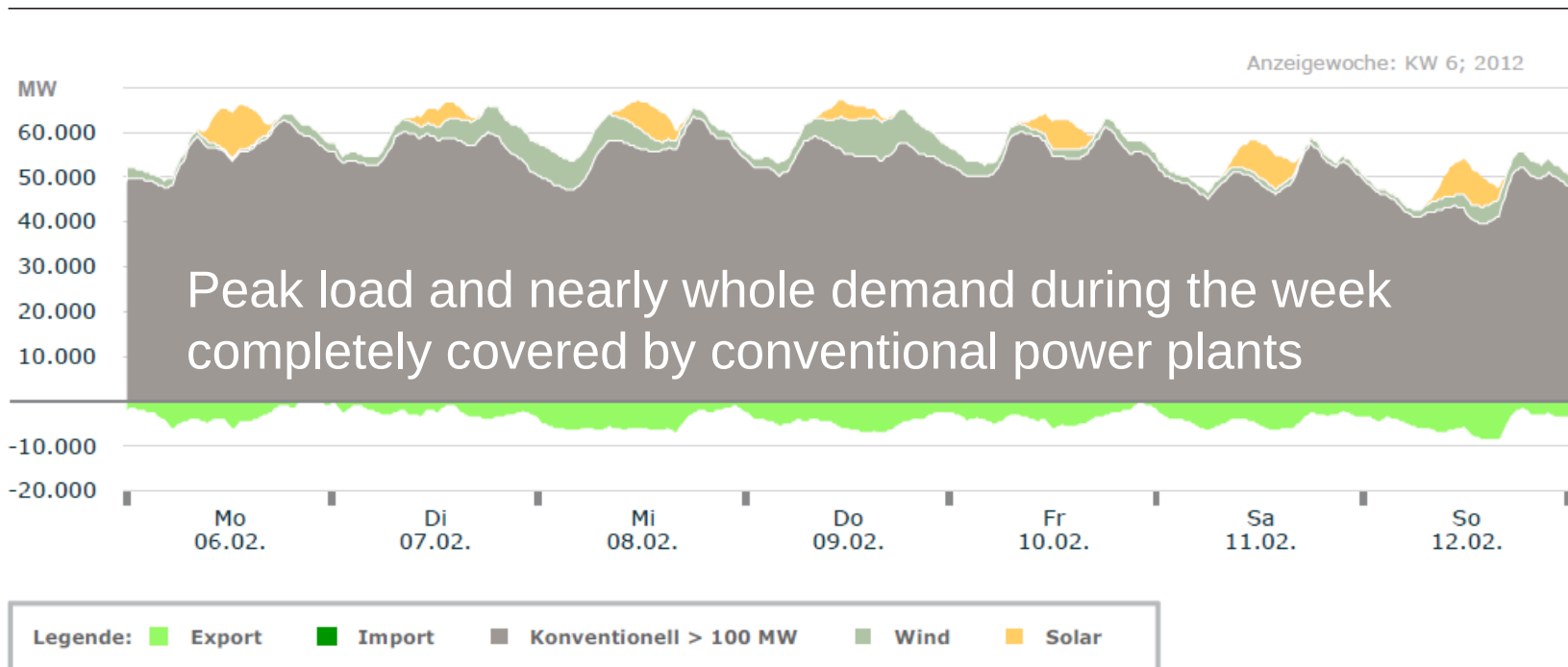


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RES challenges

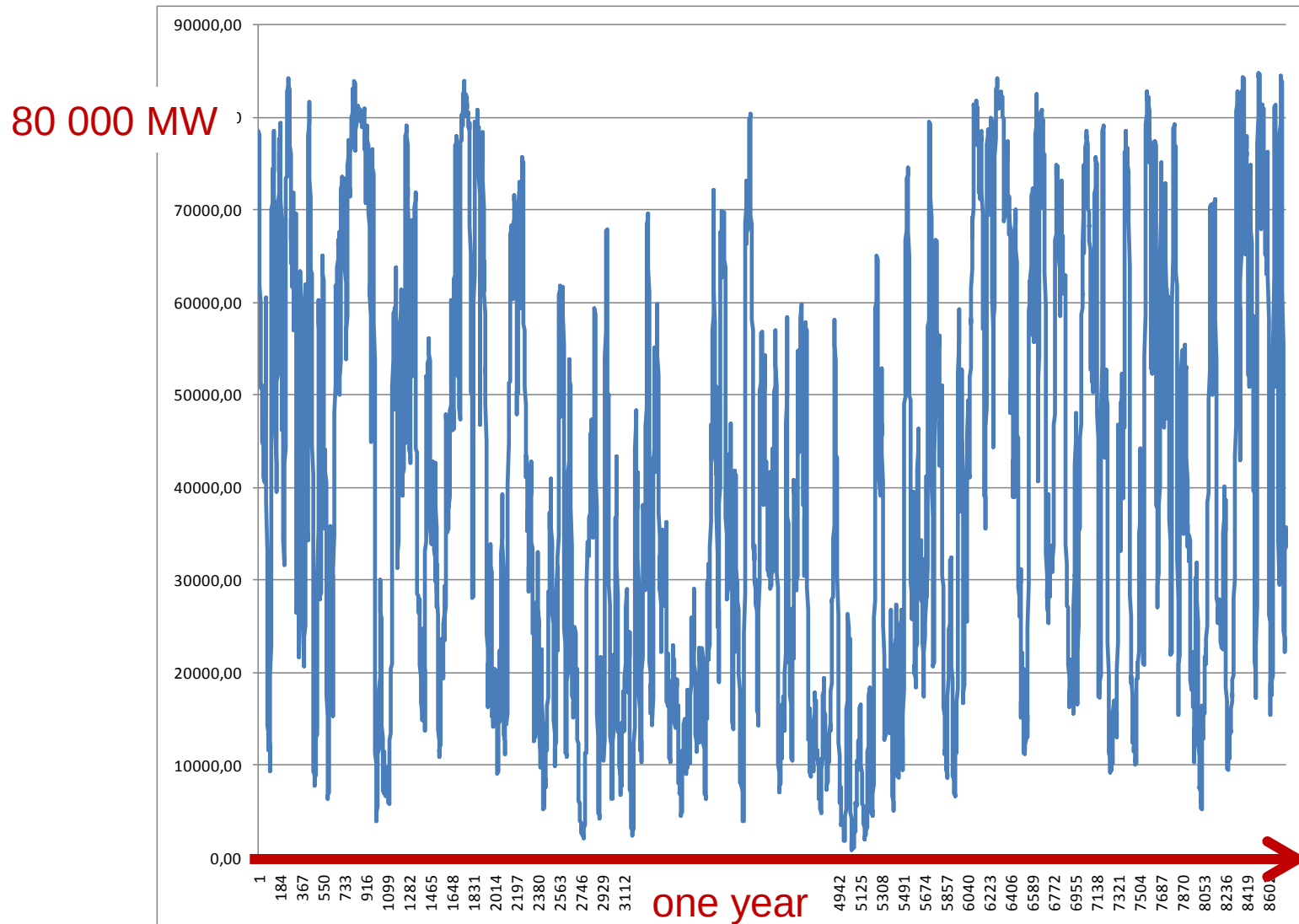
No production from RES



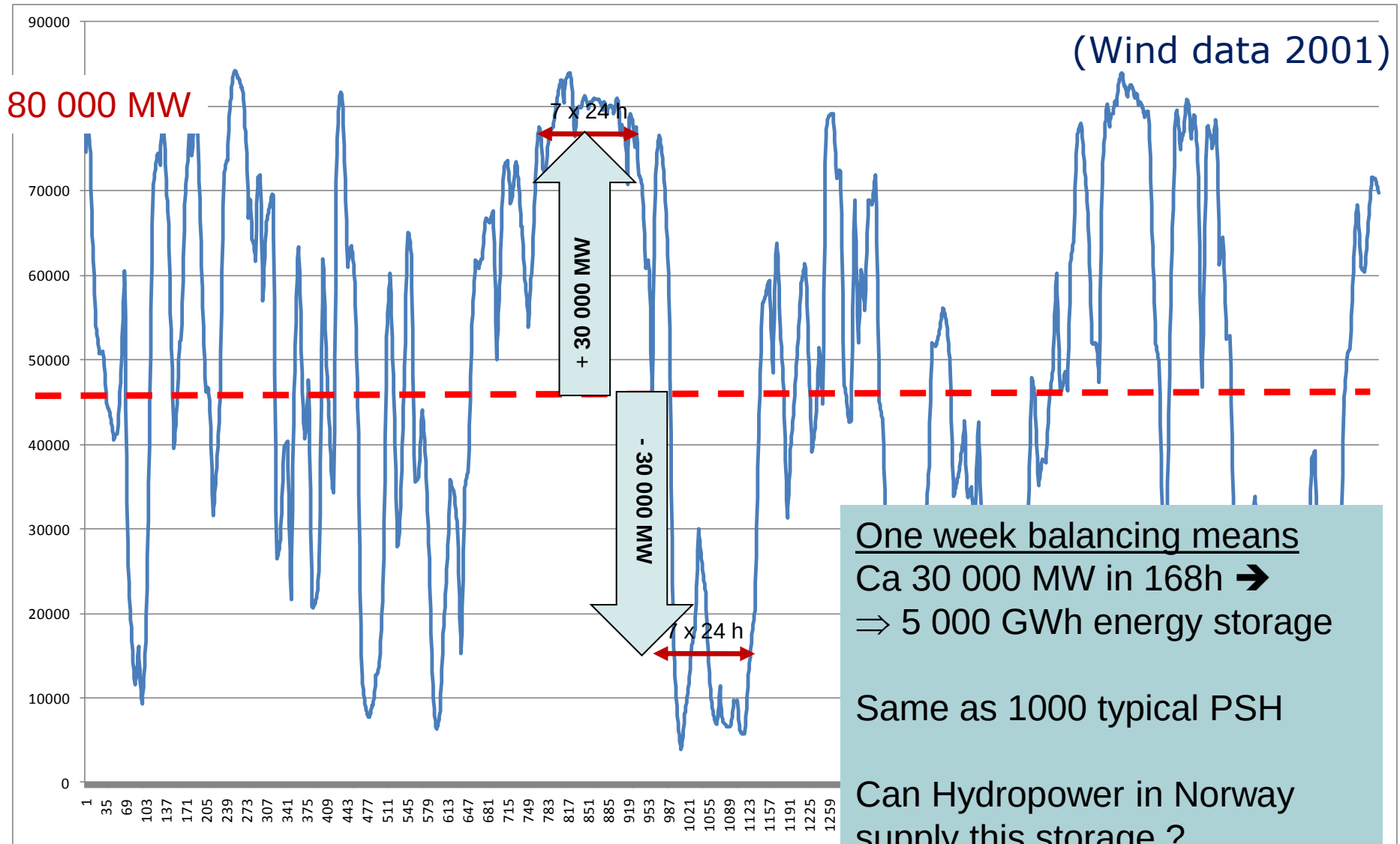
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URL: <http://www.ise.fraunhofer.de/en/renewable-energy-data>

Simulated wind production in the North Sea area in 2030 – 95 000 MW installed capacity



Wind Power North-Sea Region - Jan – March



One week balancing means
Ca 30 000 MW in 168h →
⇒ 5 000 GWh energy storage

Same as 1000 typical PSH

Can Hydropower in Norway
supply this storage ?

Hydropower in Norway – Resource base

Water, high head

Large natural reservoirs

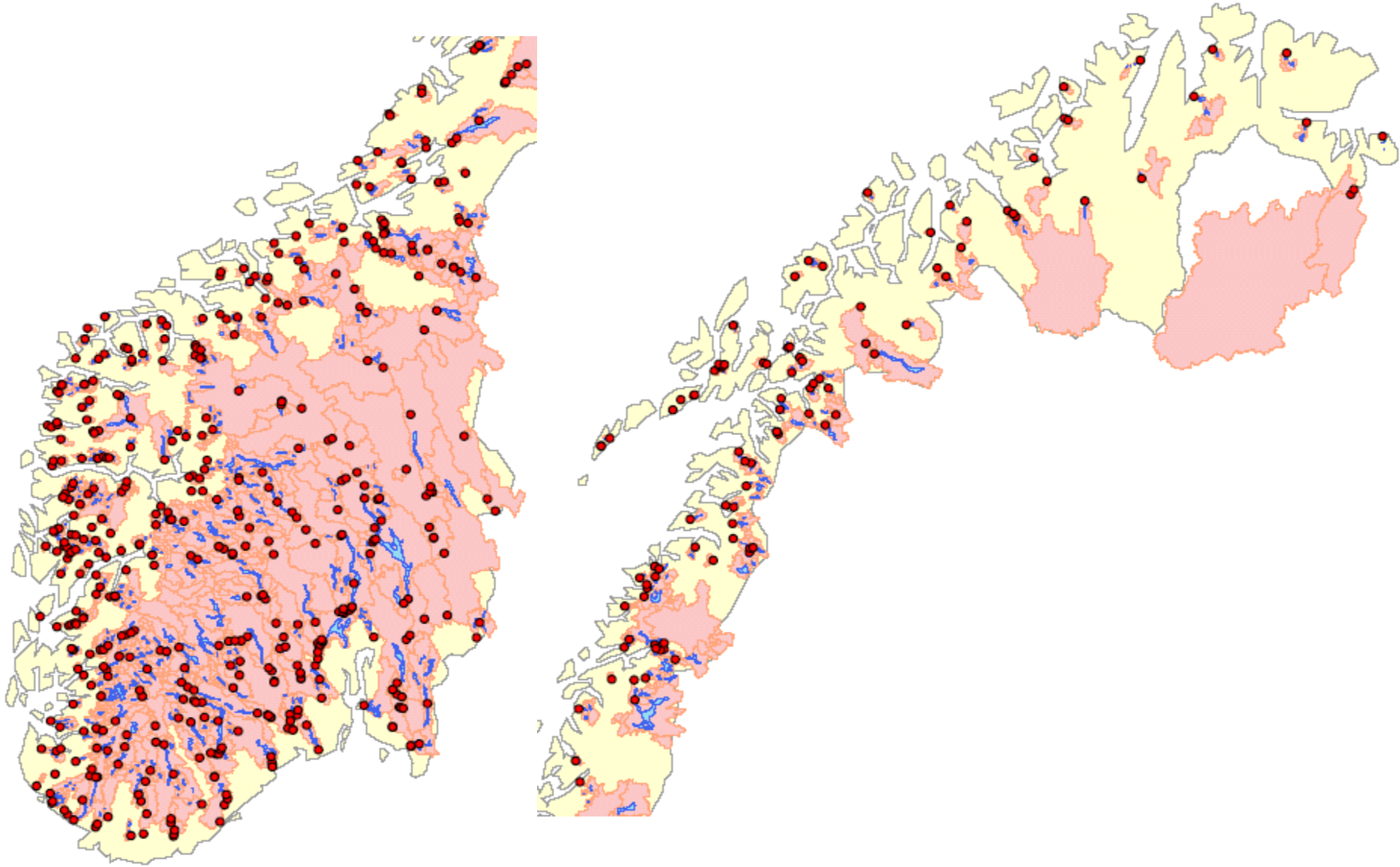


Norwegian hydropower



Solid rocks providing great opportunities to hide penstock and power plants inside the mountains

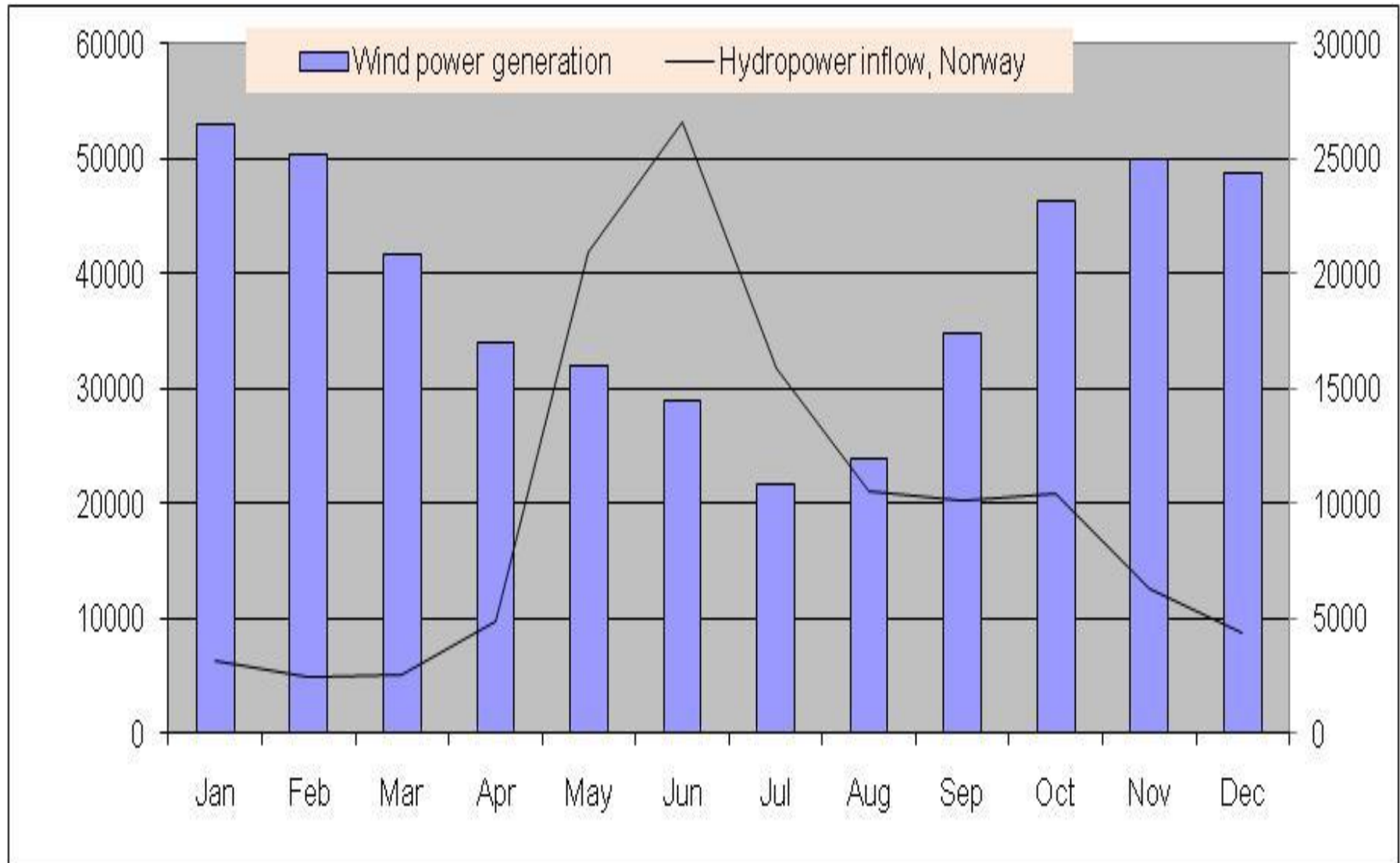
333 Large hydropower stations in Norway



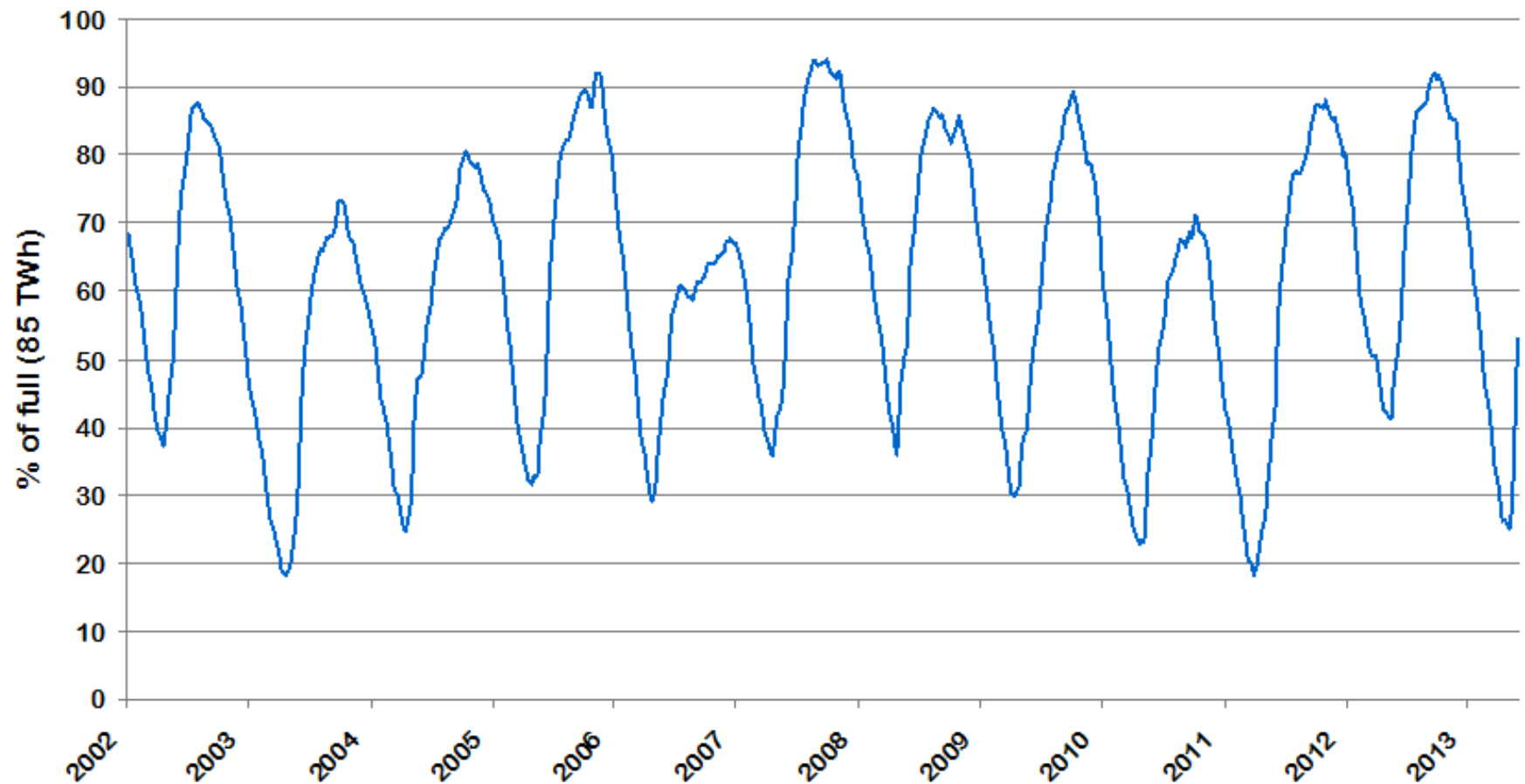


Blåsjø 7.8TWh RESERVOIR (1000 times Goldistal)

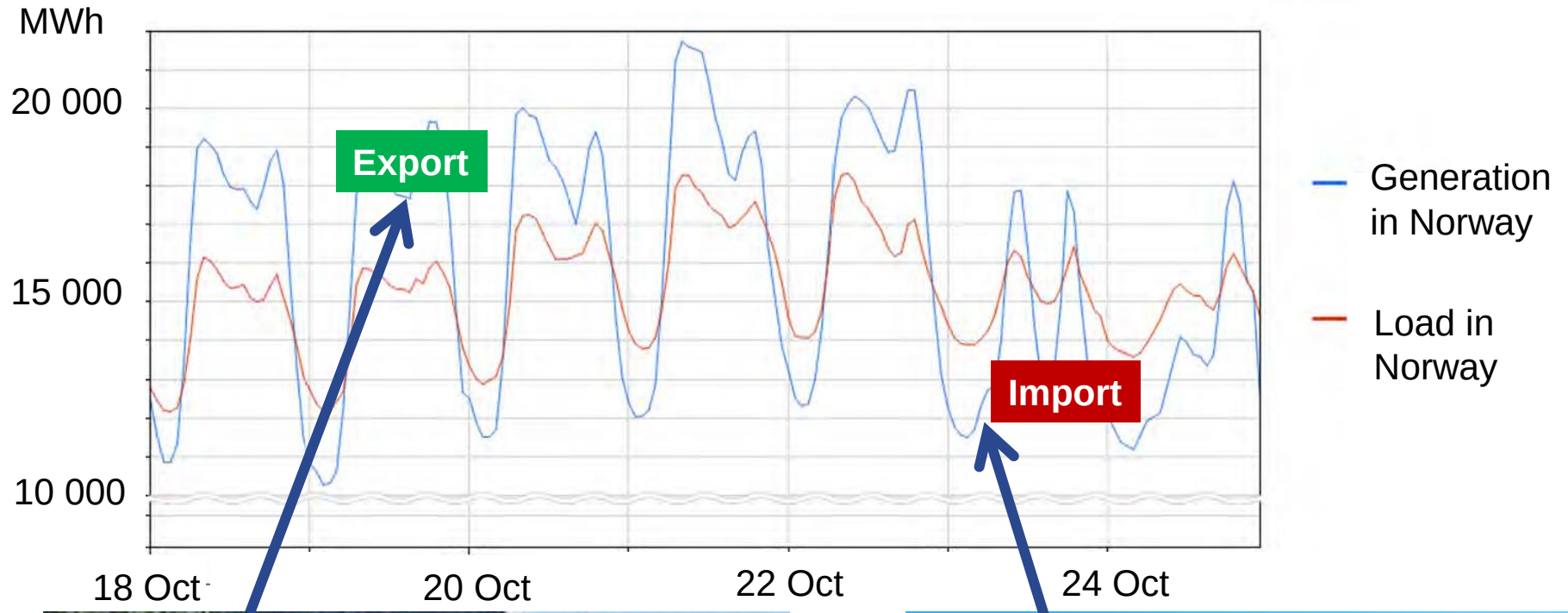
Wind and hydropower seems like a good match



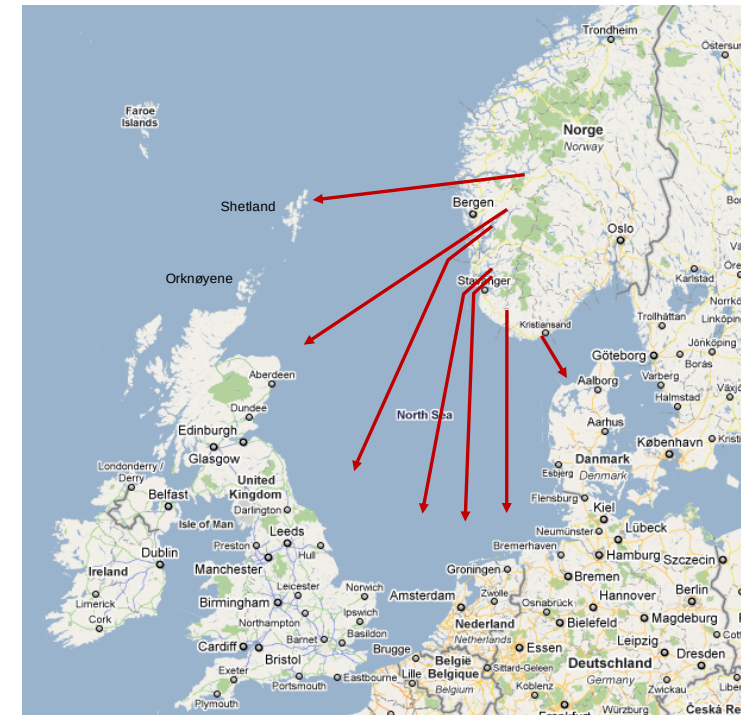
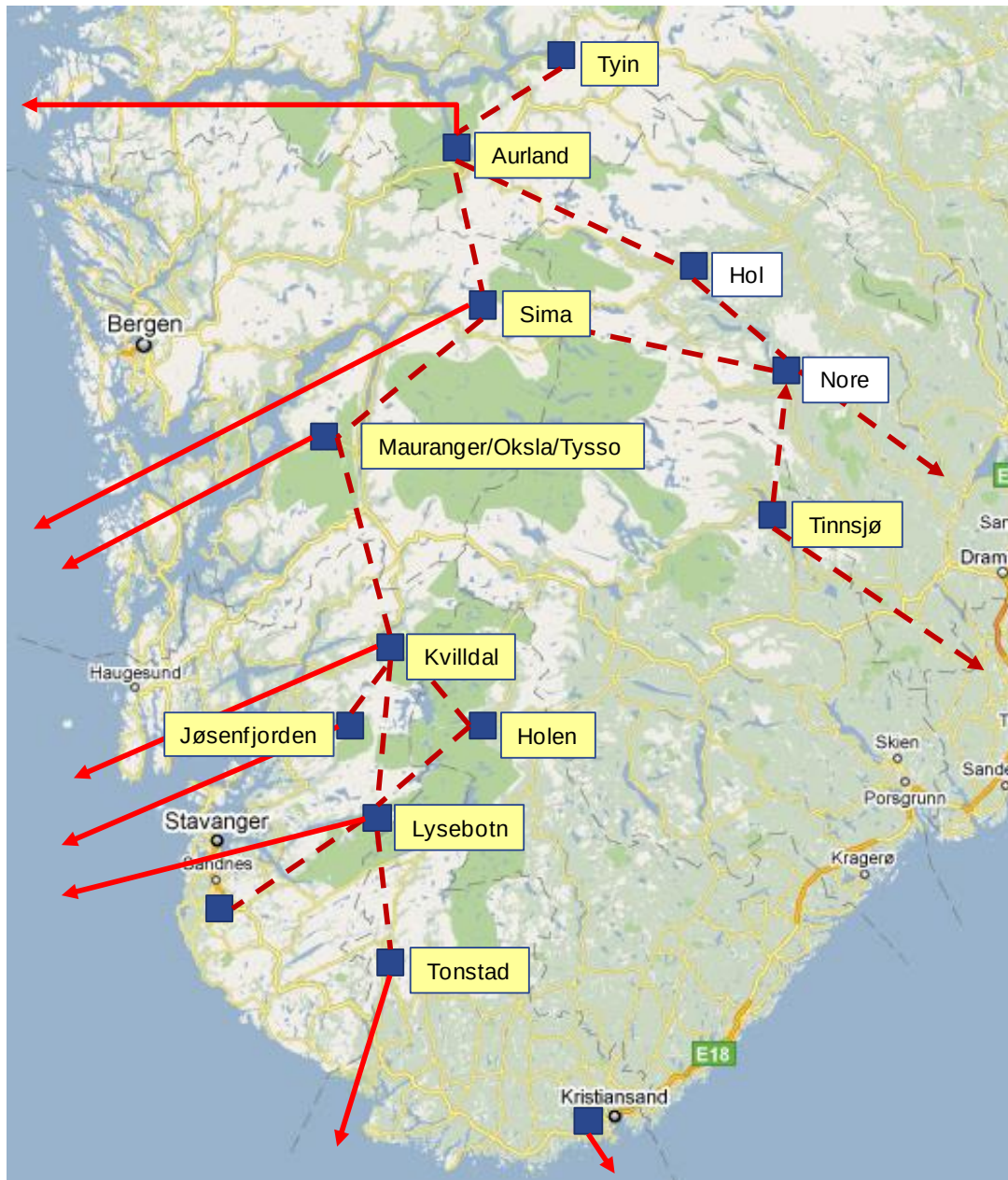
Energy content (%) in Norwegian hydropower reservoirs (2002-2013)



Norwegian hydro and Danish wind



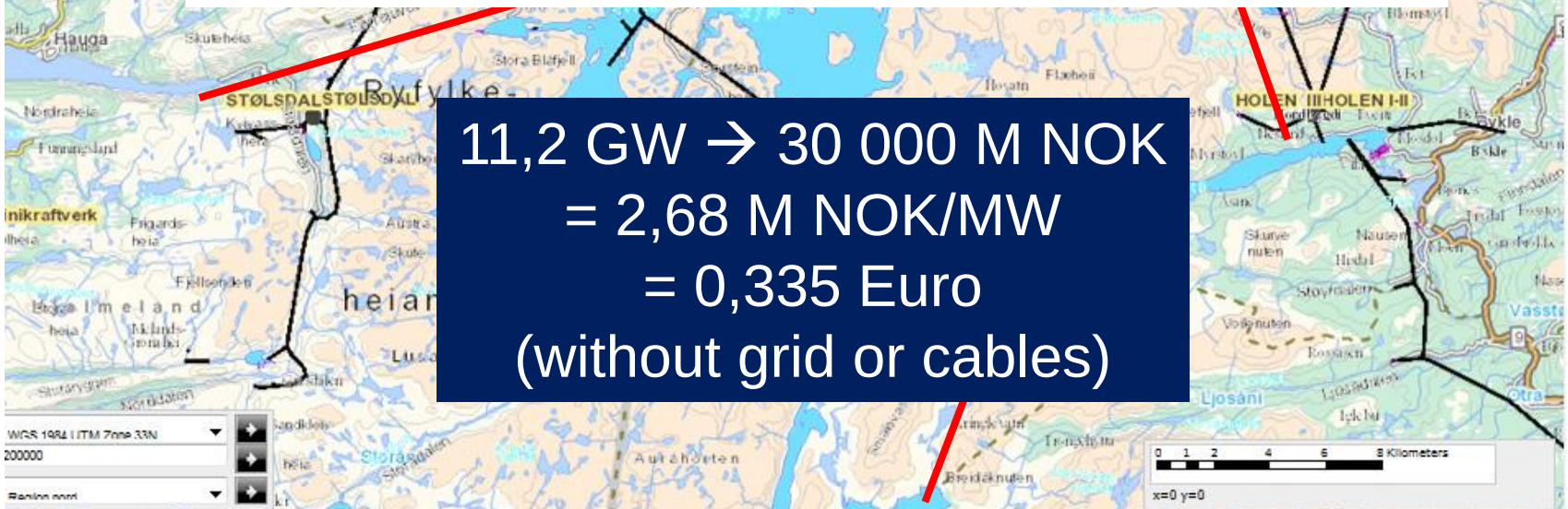
CEDREN Case study 2030



The technical potential



20 000 MW in southern Norway possible



11,2 GW → 30 000 M NOK
= 2,68 M NOK/MW
= 0,335 Euro
(without grid or cables)

Environmental impacts



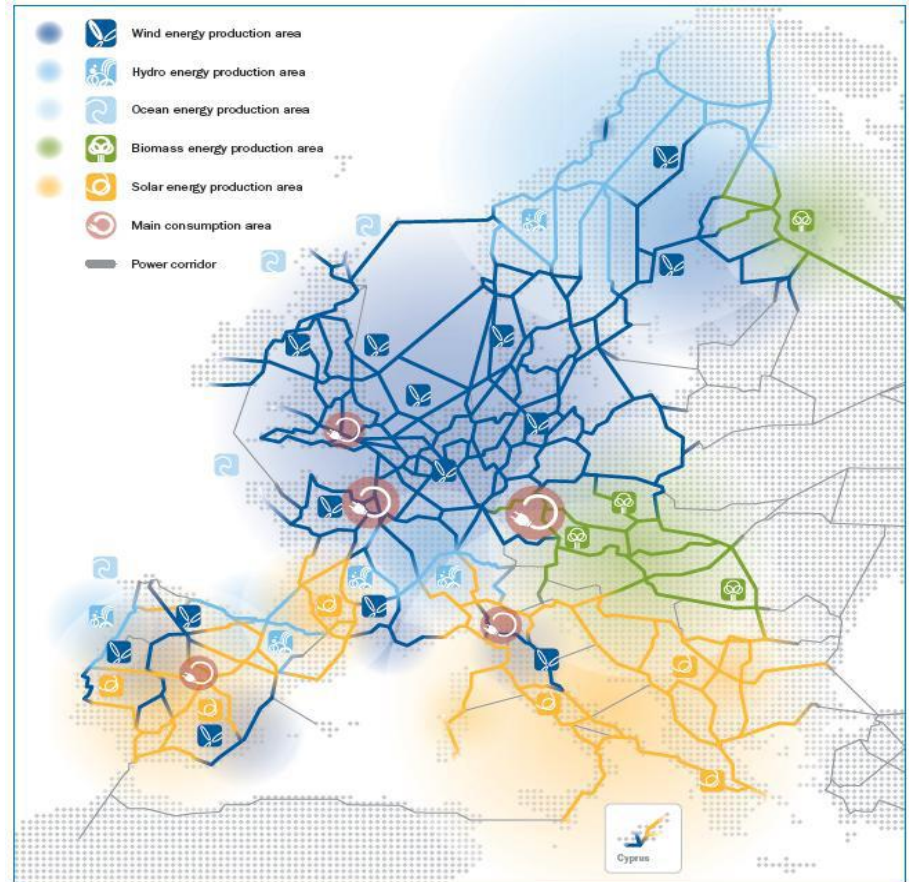
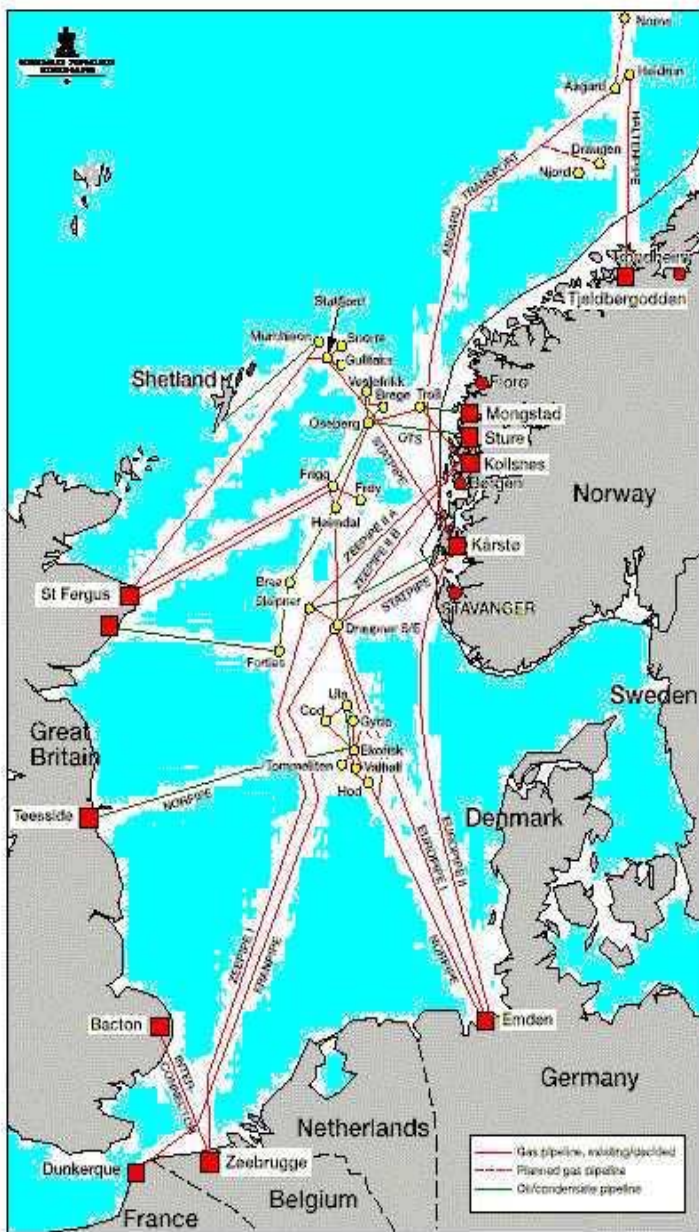
Social acceptance





← Natural gas grid today

Can we establish a similar *electricity* grid for exchange?

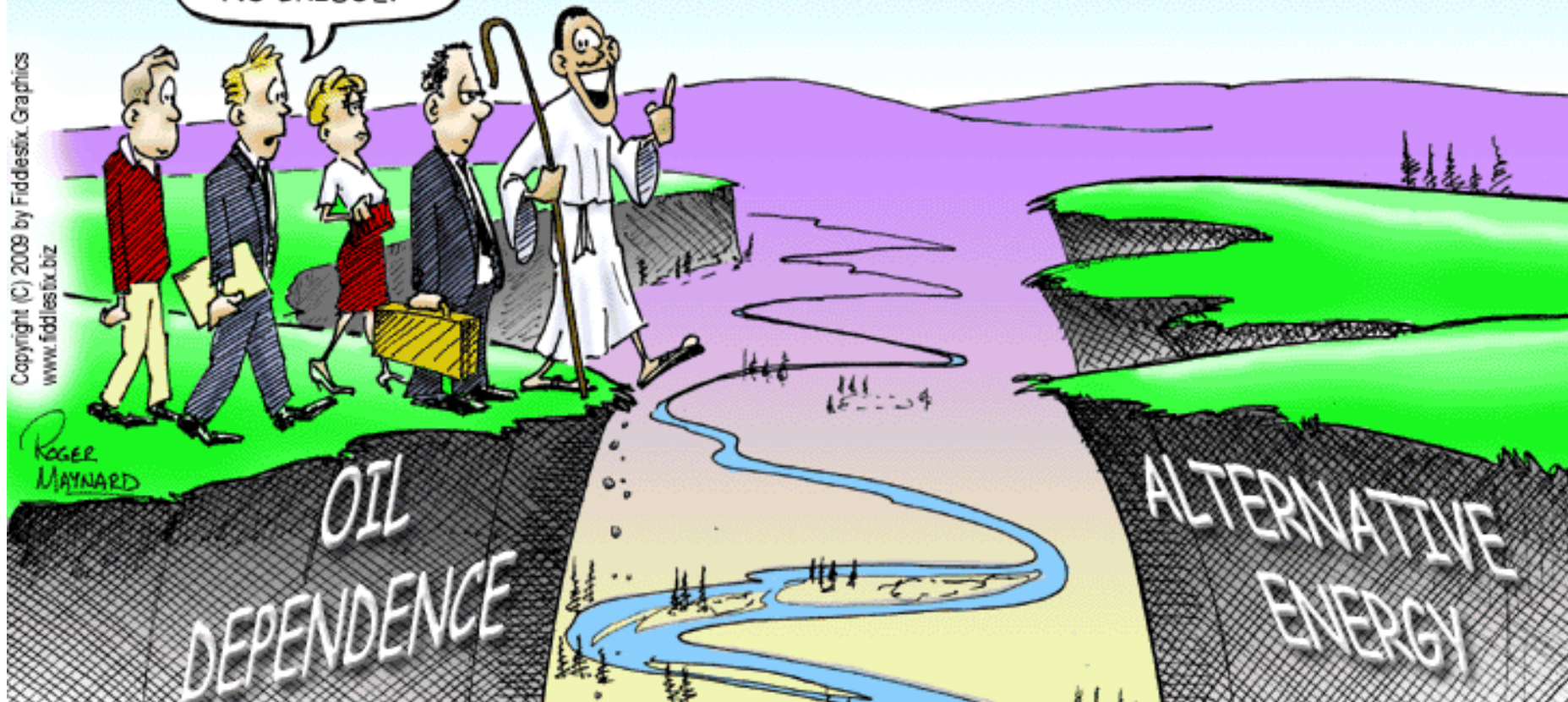


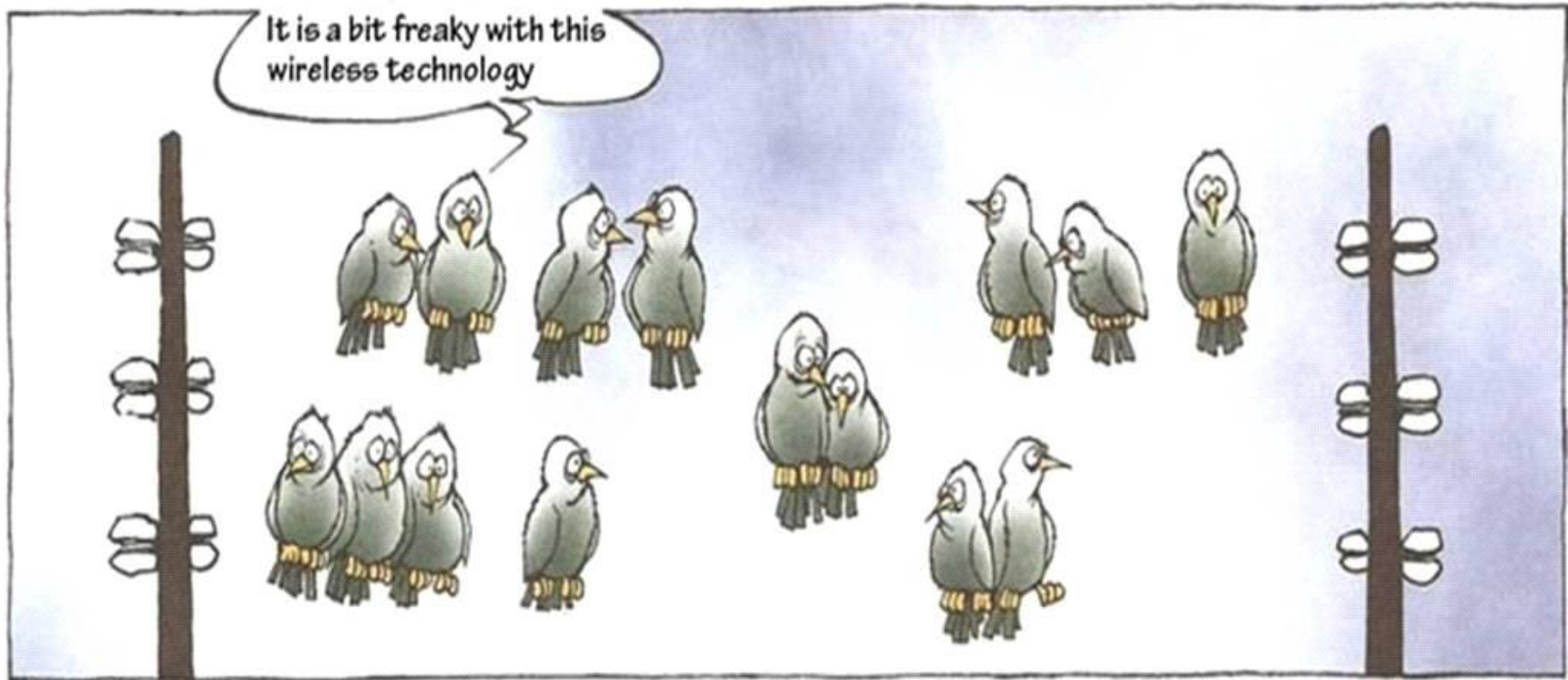
PRESIDENT OBAMA'S ENERGY PLAN:

FOLLOW ME TO THE PROMISED LAND OF
CLEAN, RENEWABLE ENERGY AND
INDEPENDENCE FROM FOREIGN OIL.

BUT THERE'S
NO BRIDGE.

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- Uncertain future – many scenarios
- Rapid changes may come (...Fukushima)
- Hydro reservoirs = always an excellent energy storage
- We probably need governmental agreements and new markets
- Norwegian hydro: Large opportunity for Europe